

Implementation Research Report

Distributed Brain Connectivity Analysis for Schizophrenia

RQFM (Riemannian Quantum Feature Map) + PQFL (Personalized Quantum Federated Learning)

Subproject 6 | Federated QML + fMRI | June 2025

Component	Recommended Framework	Version	Rating
Quantum ML	PennyLane	0.45.0	5/5
Federated Learning	Flower (flwr)	1.23+	5/5
Riemannian Geometry	geomstats + pyRiemann	2.8.0 / 0.11	4.5/5
fMRI Processing	nilearn + fMRIPrep	0.13.1 / 25.2.5	5/5
Total Subjects Available	~4,886 (open) to ~5,890 (restricted)	7 sites	-

Table 1: Executive Summary of Key Research Findings

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1. Quantum ML Frameworks for RQFM

The Riemannian Quantum Feature Map (RQFM) requires a QML framework capable of custom quantum circuit construction, flexible feature map design, automatic differentiation, and integration with classical deep learning frameworks. After comprehensive evaluation of five major QML platforms, **PennyLane emerges as the clear primary recommendation** for RQFM implementation, with Qiskit Machine Learning as a viable alternative for kernel-based approaches. TensorFlow Quantum is not recommended due to low maintenance status.

1.1 Framework Comparison

Framework	Version	RQFM Suitability	Maintenance	PyTorch Integration
PennyLane	0.45.0	Best (5/5)	Very Active	Native (TorchLayer)
Qiskit ML	0.9.0	Good (4/5)	Active (community)	Via connector
TF Quantum	0.7.2	Fair (2/5)	Low activity	N/A (TF native)
Cirq + recirq	1.4.1	Fair (2/5)	Google-maintained	None
Amazon Braket	1.115.0	Fair (3/5)	Active	Via PL plugin

Table 2: QML Framework Comparison for RQFM Implementation

1.2 PennyLane (Primary Recommendation)

PennyLane, developed by Xanadu Quantum Technologies, provides the most flexible and well-maintained platform for implementing custom quantum feature maps. Its core paradigm of "quantum differentiable programming" treats quantum circuits as neural network layers, enabling seamless hybrid classical-quantum model construction. The library requires Python 3.11 or later and supports over 30 quantum hardware backends for future deployment.

Key capabilities for RQFM: (1) Custom feature maps via arbitrary gate composition; (2) Native `qml.kernels` module for quantum kernel methods; (3) `qml.qnn.TorchLayer` for native PyTorch integration; (4) `lightning.qubit` C++ simulator achieving approximately 10x speedup over default.qubit; (5) Multiple differentiation methods (parameter-shift, adjoint, backprop); (6) Batched circuit execution for efficient training.

Installation: `pip install pennylane pennylane-lightning torch geomstats scikit-learn`

The recommended RQFM architecture follows a multi-stage pipeline: fMRI data is first processed into functional connectivity covariance matrices (SPD matrices), which are then mapped to tangent space via log-map at the Frechet mean using geomstats. The resulting tangent vectors undergo PCA dimensionality reduction (5050 dims to 64-256 dims for Schaefer 100 ROI), followed by a trainable classical encoder reducing to `n_qubits` dimensions. Angle embedding then maps these features into quantum Hilbert space, where `StronglyEntanglingLayers` provide the variational classification circuit.

1.3 Critical Academic Finding: SPD + Quantum Connection

Two papers provide direct precedents for the RQFM concept. **Alavi et al. (2024), arXiv:2406.04055** demonstrates a complete pipeline: polynomial feature expansion, SPD matrix generation, eigenvector decomposition, normalization, and quantum state preparation using AngleEmbedding + StronglyEntanglingLayers in PennyLane. Their SPD-Enhanced Hybrid model achieved MSE 0.00031 and R-squared of 0.98765. The key insight is that "SPD matrices, which belong to the Riemannian manifold geometrical space, are similar to quantum states," ensuring data representation alignment with the quantum computational framework.

Alavi et al. (2025), arXiv:2504.06328 extends this by unifying QML as a geometry-centric extension of classical geometric ML. It establishes mathematical parallels between SPD manifolds (Sym+(n)) and quantum state manifolds (projective Hilbert spaces), and bridges information geometry with quantum Fisher information. **Importantly, no existing work combines quantum feature maps with SPD Riemannian geometry specifically for fMRI functional connectivity classification**, confirming the research gap that our RQFM addresses.

2. Federated Learning Frameworks for PQFL

Personalized Quantum Federated Learning (PQFL) requires a federated learning framework that supports custom aggregation strategies, parameter splitting for personalization, differential privacy, and crucially, integration with quantum variational circuits. After evaluating five major FL frameworks, **Flower (flwr) is the clear winner** as it is the only framework with official Quantum Federated Learning support via PennyLane integration, added in version 1.23 (November 2025).

Framework	Verdict	QML Support	Personalization	Medical Imaging
Flower	USE (primary)	Official PL integration	FedPer/FedRep baselines	MONAI integration
PySyft	Skip	None	None	No
NVIDIA FLARE	Secondary	None	Limited	Best (MONAI native)
FedML	Skip	None	Limited	No
Substra	Niche	None	None	EU healthcare

Table 3: Federated Learning Framework Comparison

2.1 Flower + PennyLane Integration

The Flower + PennyLane integration is confirmed working through the official `quickstart-pennylane` example released in November 2025. The integration follows a clear pattern: define a `QNode` with PennyLane, wrap it with `qml.qnn.TorchLayer` to convert it into an `nn.Module`, compose with other PyTorch layers for the full hybrid model, train using standard PyTorch optimizers, and extract parameters as numpy arrays for Flower communication. Flower aggregates the numpy arrays via FedAvg and returns the updated global parameters. The quantum circuit structure (ansatz type, number of qubits, entanglement pattern) is fixed across all sites, and only the continuous parameters of the variational layers are federated.

FedPer Implementation: Following Arivazhagan et al. (2019), we split model parameters into shared layers (classical encoder + VQC base layers, federated via FedAvg weighted average) and personalization layers (VQC personalization layers + classification head, kept local at each site). This is implemented in Flower by overriding `get_parameters()` and `set_parameters()` in the `NumPyClient` to filter out personalization parameters.

2.2 Riemannian-Aware Federated Aggregation

Standard FedAvg computes arithmetic mean of parameters, which violates the geometric structure of SPD manifold data. The **FedSPDnet paper (arXiv:2604.22494, April 2026)** proposes two geometry-aware federated aggregation strategies specifically for SPD network architectures. **ProjAvg (Projected Averaging)** computes the arithmetic mean of Stiefel-valued parameters and projects back onto the Stiefel manifold via polar decomposition (SVD). This is simpler and more robust. **RLAvg (Retraction-Lifting Averaging)** maps each local weight to the tangent space at a reference point, computes the Euclidean mean in tangent space, and retracts back to the manifold. No official code release exists for either strategy, requiring custom implementation. For our PQFL architecture, the primary VQC parameters being federated are classical (not Stiefel-valued), so standard FedAvg is appropriate for those, while the Frechet mean should be used for any SPD matrix aggregation.

3. fMRI Processing Pipeline and Dataset Access

The fMRI processing pipeline transforms raw neuroimaging data into functional connectivity matrices suitable for Riemannian analysis. The recommended pipeline follows a two-stage approach: offline preprocessing with fMRIPrep followed by online feature extraction with nilearn. All preprocessing is performed once and cached; only the model forward/backward pass and federated communication happen during training.

3.1 Processing Tools

Tool	Version	Role	Key Function
fMRIPrep	25.2.5	Preprocessing	BOLD reference, motion correction, SDC, normalization
nilearn	0.13.1	Feature extraction	NiftiLabelsMasker, ConnectivityMeasure
nibabel	5.4.2	NIfTI I/O	Load/save fMRI data in NIfTI format
templateflow	25.1.2	Brain templates	MNI152NLin2009cAsym template access
neurocombat	0.2.12	Harmonization	ComBat site-effect removal on FC features
bctpy	0.6.1	Graph metrics	Modularity, clustering, efficiency on FC matrices

Table 4: fMRI Processing Tool Stack

Key pipeline: fMRIPrep outputs (preprocessed BOLD + confounds) are loaded by nilearn, where `NiftiLabelsMasker` extracts ROI time series using the Schaefer 2018 atlas (100/200/400 ROIs), applies confound regression and band-pass filtering. `ConnectivityMeasure` then computes functional connectivity matrices, with the "tangent" kind being particularly valuable for multi-site analysis as it projects covariance matrices to a vector space. ComBat harmonization is applied to remove site effects from the tangent

space representations while preserving biological variance (diagnosis, age, sex).

3.2 Dataset Access and Characteristics

Dataset	Subjects	Access	Format	Sites
COBRE	~147	Open (NITRC)	Raw NiftI	1
FBIRN	~473	DUA (NITRC)	Raw DICOM/NiftI	7
MCIC	~247	Free (COINS)	Raw NiftI	4
SRPBS-1600	~1,410 open	Registration (Synapse)	Raw + pre-computed FC	11 (Japan)
LA5c/CNP	~265	Open (OpenNeuro)	BIDS raw	1 (UCLA)
BSNIP-2	~2,103	NDA controlled	NDA format	5+
TCP 2025	~241	Open (OpenNeuro)	BIDS raw	1 (Melbourne)

Table 5: Seven-Site Federated Dataset Strategy

The total available subjects across all seven sites range from approximately 4,886 with open access to approximately 5,890 when including restricted access datasets. The five training sites (COBRE, FBIRN, MCIC, SRPBS-1600, LA5c/CNP) provide diverse multi-site data with different scanners and protocols, while the two validation sites (BSNIP-2, TCP 2025) serve as external held-out test sets to evaluate cross-site generalization. Notably, the SRPBS Multi-disorder Connectivity Dataset provides pre-computed resting-state functional connectivity matrices, making it immediately usable for federated analysis without preprocessing. The LA5c/CNP and TCP datasets are already in BIDS format on OpenNeuro with CC0 licensing, making them the easiest to integrate. BSNIP-2 requires NIMH Data Archive controlled access with IRB verification, making it the most restrictive but also the most phenotypically rich dataset with biotyping information.

4. Riemannian Geometry Libraries for SPD Manifold Operations

Functional connectivity matrices derived from fMRI are symmetric positive definite (SPD) matrices that naturally reside on a Riemannian manifold rather than in Euclidean space. Applying standard Euclidean operations (arithmetic mean, linear PCA) to SPD matrices violates their geometric structure, leading to suboptimal classification performance. The recommended stack uses geomstats for core Riemannian geometry operations (log/exp maps, parallel transport, Frechet mean), pyRiemann for covariance estimation and tangent-space ML pipelines, and optionally spd_learn for differentiable PyTorch-based deep learning on SPD manifolds.

Library	Version	Backend	Log/Exp Map	Frechet Mean	Parallel Transport
geomstats	2.8.0	NumPy/PyTorch	Yes (closed-form AIRM)	Yes (FrechetMean class)	Yes (closed-form)
pyRiemann	0.11	NumPy only	Via tangent_space()	Yes (mean_riemann)	No

spd_learn	0.1.0	PyTorch (GPU)	Yes (differentiable)	No	Yes (differentiable)
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Table 6: Riemannian Geometry Library Comparison

4.1 Key Operations and Performance

For the affine-invariant Riemannian metric (AIRM), the canonical metric for SPD matrices, all core operations have closed-form solutions in geomstats. The geodesic distance between two SPD matrices P and Q is computed as $d(P,Q) = \|\log(P^{-1/2} Q P^{-1/2})\|_F$. The log-map projects an SPD matrix to the tangent space at a reference point (typically the Frechet mean), while the exp-map performs the inverse operation. Crucially, parallel transport under AIRM has a closed-form solution via congruence transform: $S' = E * S * E^T$, where $E = (\text{target} * \text{base}^{-1})^{1/2}$. This is essential for federated aggregation where updates from different sites must be transported to a common reference point.

Operation	100 x 100	200 x 200	400 x 400
Eigendecomposition	~1 ms	~8 ms	~60 ms
Log-map / Exp-map	~2 ms	~15 ms	~100 ms
Geodesic distance	~1.5 ms	~10 ms	~80 ms
Parallel transport (AIRM)	~5 ms	~30 ms	~200 ms
Frechet mean (50 samples, 10 iters)	~100 ms	~800 ms	~5000 ms

Table 7: SPD Operation Performance (CPU, approximate)

All operations are well within standard workstation capabilities. For 100 ROI matrices (Schaefer parcellation), the Frechet mean of 50 matrices computes in approximately 100ms. The critical bottleneck is eigendecomposition ($O(n^3)$), but even at 400 ROIs, per-matrix operations remain under 200ms. GPU acceleration via the PyTorch backend provides 2-10x speedup for batched operations, though some operations fall back to CPU due to `scipy.linalg.eigh` dependencies in certain code paths.

4.2 Quantum Feature Mapping from Tangent Space

The connection between tangent space projection and quantum feature maps is established by Vlastic (2025, arXiv:2509.02795), which proves that geodesics on the embedded manifold of quantum operators are also geodesics on the codomain of the quantum feature map. This means the Riemannian structure of the SPD manifold is preserved under quantum encoding, validating our RQFM approach. The practical implementation follows: (1) compute tangent vectors at the Frechet mean using log-map; (2) vectorize the symmetric tangent matrix by extracting the upper triangular elements ($n(n+1)/2$ dimensions); (3) apply tangent PCA to reduce from 5050 to 256 dimensions; (4) pass through a trainable classical encoder to n_{qubits} dimensions; (5) apply angle encoding via RY rotations into the quantum Hilbert space.

5. Integration Architecture Blueprint

The integration of Quantum ML, Federated Learning, Riemannian Geometry, and fMRI processing follows a **layered hybrid quantum-classical** design where the quantum layer is inserted at the "bottleneck" of the model, after classical dimensionality reduction of tangent vectors and before final classification. This keeps

qubit requirements feasible (8-16 qubits) while leveraging quantum expressivity for the most critical feature transformation. The architecture consists of a central Flower server performing FedPer aggregation of shared parameters, with 7 site-specific Flower clients each running the complete hybrid VQC pipeline locally. Personalization occurs at two levels: site-specific classical pre-processing (harmonization) and site-specific VQC personalization layers that are never communicated to the server.

5.1 End-to-End Data Flow

The data flow proceeds through six stages, with stages 1-3 performed offline and cached: **Stage 1 (fMRI Pre-processing)**: fMRIPrep outputs are loaded by Nilearn, confound regression and band-pass filtering are applied, and ROI time series are extracted using Schaefer 2018 parcellation. **Stage 2 (FC Matrix Construction)**: Pearson correlation produces $n \times n$ SPD matrices with regularization ($C + \lambda I$, $\lambda = 1e-3$) to ensure positive definiteness. **Stage 3 (Riemannian Processing)**: Frechet mean is computed as the reference point, log-map projects SPD matrices to tangent vectors at the Frechet mean (dimension $n(n+1)/2$), and optional ComBat harmonization removes site effects in tangent space. **Stage 4 (Classical Dimensionality Reduction)**: Tangent PCA reduces 5050 to 256 dimensions, then a trainable classical encoder (nn.Linear layers with BatchNorm, GELU, and Tanh for quantum-compatible clamping) reduces to n_{qubits} dimensions. **Stage 5 (RQFM Quantum Feature Map + VQC)**: Angle encoding maps n_{qubits} features to n_{qubits} qubits via RY rotations, StronglyEntanglingLayers (3 base layers, shared) and BasicEntanglerLayers (1 personalization layer, local) provide the variational circuit, and Pauli-Z expectation values produce n_{qubits} classical outputs. **Stage 6 (Classification Head)**: A local nn.Linear layer maps quantum outputs to 2-class probabilities (SZ vs HC).

5.2 Federated Communication Protocol

Each federated round follows the Flower protocol: (1) The server broadcasts the current shared parameters (classical encoder + VQC base layer weights) to all participating clients; (2) Each client sets the shared parameters while retaining its personal parameters (VQC personalization layers + classification head), then performs local training for E epochs (default 5) using AdamW optimizer with cosine annealing learning rate scheduling; (3) Each client returns only the updated shared parameters and sample count to the server; (4) The server aggregates shared parameters via weighted FedAvg (proportional to sample sizes) and distributes the new global parameters. Personalization parameters never leave the client site, ensuring site-specific adaptation while benefiting from global knowledge sharing. The recommended configuration is 50 communication rounds with all 7 sites participating each round ($\text{fraction_fit}=1.0$).

5.3 Non-IID Mitigation Strategy

Multi-site medical data exhibits multiple sources of non-IID distribution shift. Our four-layer mitigation strategy addresses each source: **Layer 1 (Pre-processing Harmonization)**: ComBat harmonization in tangent space removes scanner effects while preserving biological covariates (diagnosis, age, sex). Site-specific Frechet means account for different reference points. **Layer 2 (Architecture-Level Personalization)**: FedPer splits the model into shared base layers and local personalization layers, allowing each site to maintain specialized features. **Layer 3 (Training-Level Regularization)**: Optional FedProx regularization ($\mu=0.01$) prevents local divergence from the global model. L2 regularization on personalization layers prevents

overfitting. **Layer 4 (Aggregation-Level Weighting):** Weighted FedAvg proportional to sample size ensures fair representation. Optional gradient clipping limits the impact of outlier sites with extreme updates.

6. Computational Budget and Training Estimates

The computational requirements for PQFL training are remarkably modest, owing to the efficient quantum simulation at 12 qubits and the cached offline preprocessing. For circuits with fewer than 20 qubits, CPU simulation via lightning.qubit is actually faster than GPU due to GPU overhead exceeding the benefit. The state vector for 12 qubits requires only 32 KB of memory, with each circuit execution taking approximately 0.02 seconds on CPU.

Configuration	Per-Site Local	Round Time	Total (50 rounds)	Est. Cost
12 qubits, CPU, 5 epochs	~30s	~5 min	~4 hours	~\$85
12 qubits, GPU, 5 epochs	~15s	~3 min	~2.5 hours	~\$75
16 qubits, CPU, 5 epochs	~60s	~8 min	~7 hours	~\$150
Full 7-fold CV (12q, CPU)	-	-	~28 hours	~\$300-500

Table 8: Training Time and Cost Estimates

Memory requirements for SPD operations are modest: a site with 150 subjects and n=100 ROI parcellation requires approximately 12 MB for FC matrices and 6 MB for tangent vectors. For n=400, these increase to 192 MB and 96 MB respectively, still well within standard workstation capabilities. The Frechet mean computation takes approximately 5 seconds for n=100 and 30 seconds for n=400 on CPU. The total estimated cloud budget for all experiments including full 7-fold cross-validation is \$300-500, making this project highly feasible on standard academic computing resources.

7. Implementation Roadmap

The implementation follows a five-phase, ten-week plan that builds the system incrementally, validating each component before integration. This approach minimizes risk by ensuring that classical baselines are established before introducing quantum components, and that single-site quantum models work correctly before federating across seven sites.

Phase	Timeline	Focus	Key Deliverables
1: Foundation	Week 1-2	fMRI pipeline + Riemannian engine + SPD validation + unit tests	SPD matrices, tangent vectors, Frechet mean computation
2: Quantum	Week 3-4	VQC circuits in PennyLane + HybridVQC model + classical baselines	Working VQC, tangent-space SVM baseline, MDM classifier
3: Federation	Week 5-6	Flower FedPer + PQFLClient + simulation mode + MLflow tracking	Single federated round end-to-end, experiment tracking
4: Training	Week 7-8	7-fold CV + FedAvg/FedPer/FedProx comparison + external validation	Comprehensive results, cross-site generalization metrics

5: Optimization	Week 9-10	Hyperparameter search + scale to 400 ROIs + amplitude encoding + Q-MAML	Final model, scaled parcellation, advanced personalization
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Table 9: Five-Phase Implementation Roadmap

7.1 Project Structure

The project follows a modular Python package structure organized into six core modules: **data/** (fMRI pipeline, parcellation, FC construction, PyTorch datasets, site partitioning), **riemannian/** (SPD manifold operations, tangent space projection, Frechet mean computation, tangent PCA), **harmonization/** (ComBat and CovBat site-effect removal), **quantum/** (VQC circuit definitions, RQFM encoding strategies, simulator management, quantum kernel baseline), **models/** (HybridVQC model, classical encoder, RQFM module, classification head), and **federated/** (FedPer/FedProx strategies, PQFLClient, custom aggregation, parameter split/merge utilities). Configuration management uses Hydra/OmegaConf with hierarchical YAML configs for each site, and experiment tracking integrates MLflow for global and per-site metrics logging.

7.2 Risk Assessment

Risk	Likelihood	Impact	Mitigation
Barren plateaus in VQC	Medium	High	Limit circuit depth 3-5 layers; local cost function; parameter-shift gradients
Non-IID causes divergence	Medium	Medium	FedPer + ComBat harmonization + FedProx regularization
Quantum simulation too slow	Low	Medium	lightning.qubit CPU for <20 qubits; pre-compute classical ops; batch execution
SPD regularization fails	Low	High	Adaptive lambda; validate eigenvalues > 0; shrinkage estimator fallback
ComBat removes diagnostic signal	Low	High	Preserve diagnosis as biological covariate; validate with site-specific baselines
PennyLane/Flower version incompatibility	Medium	Medium	Pin versions in pyproject.toml; test integration early

Table 10: Risk Assessment Matrix

8. Complete Installation and Version Pinning

The following commands install all required dependencies with version pinning for reproducibility. The stack has been verified for compatibility across the PennyLane + Flower + geomstats + nilearn ecosystem. Python 3.11 or later is required.

Component	Package	Version	Install Command
Quantum ML	pennylane	0.45.0	pip install pennylane==0.45.0
Quantum ML	pennylane-lightning	bundled	pip install pennylane-lightning

Federated Learning	flwr	1.23+	pip install flwr
Deep Learning	torch	2.x	pip install torch
Riemannian Geometry	geomstats	2.8.0	pip install geomstats[pytorch]
Riemannian ML	pyriemann	0.11	pip install pyriemann
fMRI Processing	nilearn	0.13.1	pip install nilearn
fMRI Preprocessing	fmripred	25.2.5	Docker: nipreps/fmripred:25.2.5
Harmonization	neurocombat	0.2.12	pip install neurocombat
General ML	scikit-learn	1.x	pip install scikit-learn
Experiment Tracking	mlflow	2.x	pip install mlflow
Configuration	hydra-core	1.3+	pip install hydra-core
Graph Metrics	bctpy	0.6.1	pip install bctpy

Table 11: Complete Dependency List with Version Pinning

This implementation research confirms the technical feasibility of the PQFL + RQFM architecture for distributed brain connectivity analysis in schizophrenia. All required frameworks are actively maintained, have confirmed integration paths (especially the Flower + PennyLane official quickstart), and operate within modest computational budgets. The key innovation gap, combining quantum feature maps with SPD Riemannian geometry for fMRI functional connectivity classification, remains unfilled in the literature, positioning this project for a novel contribution at the intersection of quantum computing, differential geometry, and computational neuroscience.